Statistical Data Analysis

Project Report on Bikes Data

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Contents

Introduction

The primary objective of this document is to provide a comprehensive statistical analysis of the bike data set provided by the company.

It also aims to provide a detailed explanation of the steps taken and assumptions undertaken during the data processing stage. Furthermore, this document is dedicated to addressing the queries made by the organization and providing thorough responses backed by justifiable reasoning.

The data set used for the analysis consists of the following variables.

Variable	Description
`ticket`	ticket type
`cost`	paid fee in euros
`month`	calendar month during which the trip was made
`location_from`	start location of the trip
`location_to`	end location of the trip
`duration`	travel time in seconds
`distance`	travel distance in meters
`assistance`	status of electric assistance (0 = disabled, 1 = enabled)
`energy_used`	energy consumed by the bike in watt-hours
`energy_collected`	energy collected by the bike in watt-hours

The assumptions taken while pre-processing the data are mentioned in each section.

Data Preparation

The data was loaded from the provided bikes.data file and the variables are as follows,

```
#checking the variable names with the documentation

df.head(5)
```

	ticket	cost	month	location_from	location_to	duration	distance	assistance	energy_used	energy_collected
0	single	0.35	9	MICROTEKNIA	PUIJONLAAKSO	411.0	2150	1	19.0	2.7
1	single	1.20	5	SATAMA	KEILANKANTA	1411.0	7130	1	53.8	15.3
2	savonia	0.00	9	TASAVALLANKATU	NEULAMÄKI	1308.0	5420	1	43.0	9.9
3	savonia	0.00	10	TORI	KAUPPAKATU	1036.0	1180	1	6.5	2.1
4	single	0.30	9	TORI	TORI	319.0	1120	1	13.7	1.2

The number of observations in the data before cleaning up is 1774.

Categorical Variables

The categorical variables in the data were checked for any invalid data such as spelling mistakes or incorrect values.

- ticket
- location_from
- location_to
- month
- assistance

```
#checking categorical variables for invalids

print('ticket: ',df.ticket.unique(),end="\n\n")
print('location_from: ',df.location_from.unique(),end="\n\n")
print('location_to: ',df.location_to.unique(),end="\n\n")
print('month: ',df.month.unique(),end="\n\n")
print('assistance: ',df.assistance.unique(),end="\n\n")
```

No such invalids were found in the data set.

```
ticket:  ['single' 'savonia' 'season']

location_from:  ['MICROTEKNIA' 'SATAMA' 'TASAVALLANKATU' 'TORI' 'NEULAMÄKI' 'KEILANKANTA'
 'PUIJONLAAKSO' 'KAUPPAKATU' 'KYS' 'PIRTTI']

location_to:  ['PUIJONLAAKSO' 'KEILANKANTA' 'NEULAMÄKI' 'KAUPPAKATU' 'TORI'
 'TASAVALLANKATU' 'MICROTEKNIA' 'SATAMA' 'PIRTTI' 'KYS']

month:  [ 9  5 10  6  8  7  4]

assistance:  [1 0]
```

Numerical Variables

First, the descriptive statistics were obtained for the numerical variables in the dataset.

- distance
- duration
- cost
- energy_used
- energy_collected

```
#checking the descriptive statistics to identify possible invalids and the behavior of each variable

num_variables=['distance','duration','cost','energy_used','energy_collected']
df[num_variables].describe()
```

The results were as follows,

	distance	duration	cost	energy_used	energy_collected
count	1774.000000	1774.000000	1774.000000	1774.000000	1774.000000
mean	2460.067644	671.323563	0.491234	17.399493	5.639290
std	2352.529305	1141.708627	4.127344	17.194463	6.379593
min	-3380.000000	2.000000	0.000000	0.000000	0.000000
25%	910.000000	245.250000	0.000000	1.425000	0.600000
50%	2030.000000	546.500000	0.050000	14.350000	3.900000
75%	3605.000000	877.750000	0.550000	26.900000	8.325000
max	20770.000000	25614.000000	100.000000	144.900000	56.400000

Observations and Proceedings

Distance

- The minimum is impossible (need to check the data for any minus values as distance cannot be negative).
- Check whether there are data rows with distance is equal to 0. As stated in the project instructions, there can be invalid data rows generated when the customers try to rent a bike through the mobile app and cancel the transaction without riding the bike. We will assume that these rows are recorded with distance =0.
- Check for the outliers.

Duration

- The minimum value is not 0. That means the invalid rows generated through the android app also contains a value for duration. We will assume that the travel time is not the ‘actual’ travel time.
- Minimum value seems impossible.
- Check for the outliers.

Cost

- Cost will be analyzed by the type of ticket.
- Maximum value is 100 while the mean is 0.491 (outliers in the data must be checked).
- Assuming the season and savonia ticket users make a one-time payment for a duration, the minimum value 0 is possible.

Energy Used

- If the energy_used value is greater than 0 then the assistance should equal to 1.
- If the energy_used value is 0 then the assistance should equal to 0.
- Check for the outliers.

Energy Collected

- There’s no information on whether the bikes can charge while the users pedal.
- Since the hypothesis mention about charging stations, it is assumed that the bikes are charged through charging stations and that there’s no technical connection between the electrical assistance feature and the energy collection.
- Check for the outliers.

Data Cleaning

Distance

As mentioned in the project instructions, it was informed that there can be invalid data rows which were generated through the Android app, when users try to rent a bike through the app and cancel it before riding the bike. Also, according to the maintenance team, there were technical issues with the bikes, which may have generated invalid values in the data set.

There were 257 rows with distance as 0 and thus it was assumed that those records are the invalids generated through the Android app. Therefore, those rows were removed from the dataset.

```
#checking number of rows where distance is 0

df[df['distance'] ==0].shape[0]
```

```
257
```

Apart from that, there were 39 rows with negative value as the distance which is impossible. It was assumed that these were generated because of the technical issues the bikes had. Even though the abstract values of the distances seem to be valid, removing them from the dataset as the technical issue is not clearly mentioned.

```
#checking number of rows where distance is a negative

df[df['distance'] <0].shape[0]
```

```
39
```

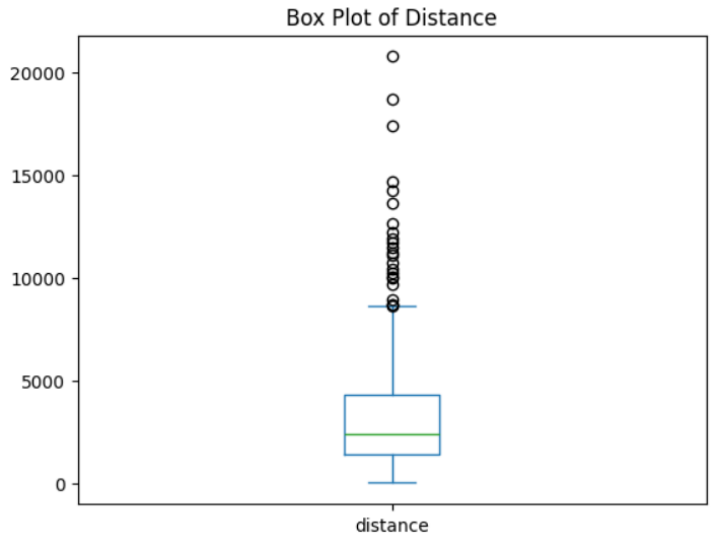
After the above invalid rows were removed, the remaining observation count was 1478.

```
#filter out the all the data with distance>0

filtered_df = df[(df['distance'] >0)]
filtered_df.shape[0]
```

```
1478
```

A box plot was created to find out about the distribution of data. By looking at diagram, it can be seen that there are outliers in the data.

```
#creating a box plot for distance data

filtered_df['distance'].plot.box()
plt.title('Box Plot of Distance')
plt.show()
```



Filtered out the observations where the distance value is greater than 10000 (see the below image).

```
filtered_df[filtered_df['distance']>10000]
```

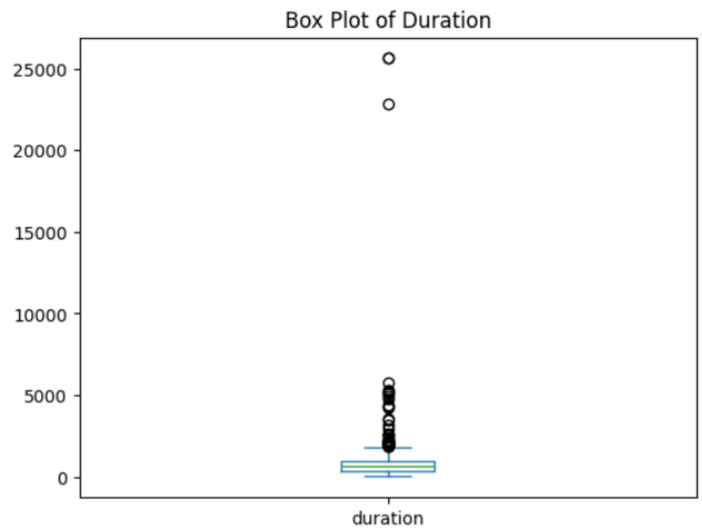
	ticket	cost	month	location_from	location_to	duration	distance	assistance	energy_used	energy_collected
214	single	1.75	6	PIRTTI	KYS	2054.0	10740	1	66.9	7.5
432	single	2.00	8	KAUPPAKATU	KAUPPAKATU	2391.0	12200	1	93.7	32.7
555	single	3.65	8	KEILANKANTA	KAUPPAKATU	4345.0	14240	1	140.6	31.8
705	single	3.55	6	PUIJONLAAKSO	PUIJONLAAKSO	4207.0	17410	1	144.9	56.4
866	single	1.70	7	SATAMA	PIRTTI	1992.0	11130	1	67.7	8.4
879	single	1.65	6	PIRTTI	KAUPPAKATU	1950.0	10040	1	47.9	9.3
1097	single	3.00	7	KAUPPAKATU	KAUPPAKATU	3545.0	13650	1	88.0	35.1
1175	single	1.75	8	PIRTTI	KAUPPAKATU	2091.0	10220	1	60.1	16.5
1202	single	4.10	8	NEULAMÄKI	NEULAMÄKI	4911.0	20770	1	124.2	52.2
1231	single	100.00	6	TASAVALLANKATU	PIRTTI	22793.0	14690	1	91.3	6.3
1430	single	4.45	6	SATAMA	SATAMA	5285.0	11440	1	71.0	24.3
1525	single	1.80	8	PIRTTI	KAUPPAKATU	2141.0	11720	1	68.9	8.1
1567	single	1.90	7	SATAMA	NEULAMÄKI	2235.0	10420	1	101.5	10.5
1649	single	4.00	5	TORI	TORI	4793.0	11200	1	98.3	55.5
1677	single	2.15	6	PIRTTI	SATAMA	2552.0	12650	1	46.0	16.5
1714	single	2.20	7	SATAMA	PIRTTI	2580.0	11880	1	63.9	23.4
1759	single	4.30	5	KEILANKANTA	KEILANKANTA	5136.0	18720	1	133.4	22.2

By looking at the data, it was decided that these outliers can be valid values and will be important in the analysis. These values could be actual data retrieved from the bikes due to long trips. These values will not be removed at this stage and if the same observations are identified as outliers considering other variables they might get removed in the cleanup process.

Duration

To visualize the data distribution, a box plot was created on duration data.

```
#creating a box plot for duration data

filtered_df['duration'].plot.box()
plt.title('Box Plot of Duration')
plt.show()
```



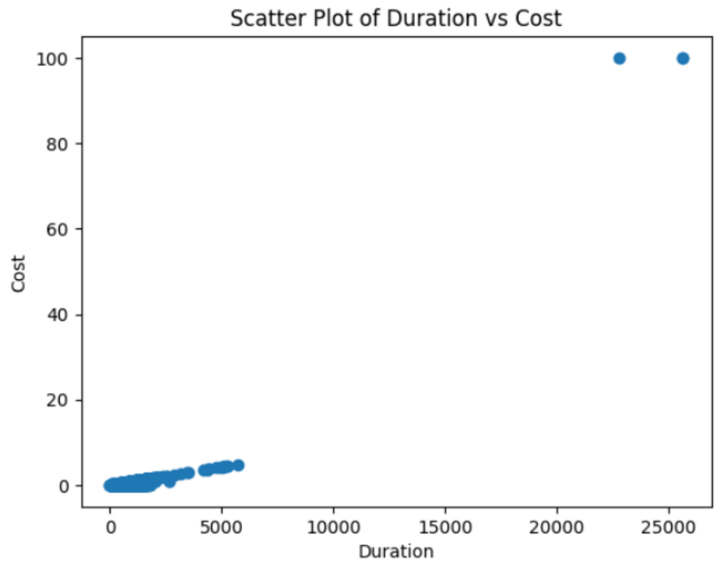
It can be seen clearly that there are some extreme outliers in the data which can be invalid. The following image contains rows where the duration value was greater than 10000.

```
filtered_df[filtered_df['duration']>10000]
```

	ticket	cost	month	location_from	location_to	duration	distance	assistance	energy_used	energy_collected
100	savonia	100.0	9	NEULAMÄKI	MICROTEKNIA	25614.0	2310	0	0.0	3.6
609	savonia	100.0	9	NEULAMÄKI	MICROTEKNIA	25614.0	2310	0	0.0	3.6
1231	single	100.0	6	TASAVALLANKATU	PIRTTI	22793.0	14690	1	91.3	6.3

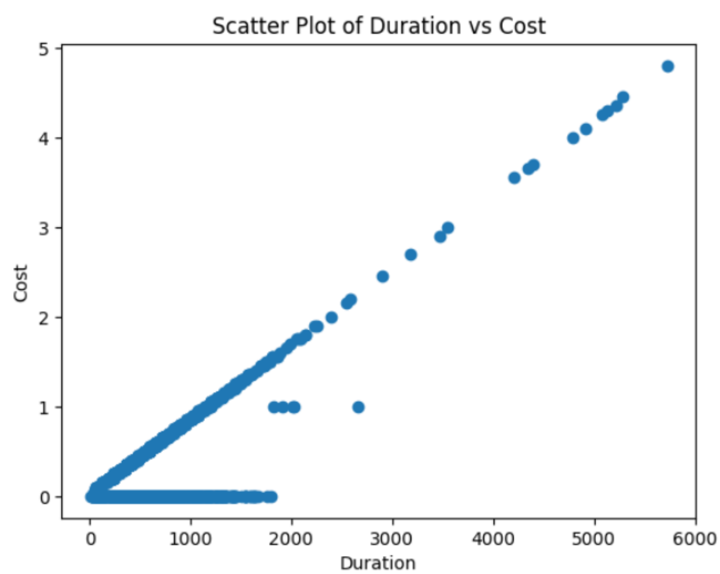
As the cost value of the above rows are 100, creating a scatter plot to see the relationship between duration and cost.

```
# Creating a scatter plot to explore the relationship between duration and cost

plt.scatter(filtered_df['duration'],filtered_df['cost'])

plt.xlabel('Duration')
plt.ylabel('Cost')
plt.title('Scatter Plot of Duration vs Cost')

plt.show()
```



The above plot shows that the rows where the duration > 10000 are also outliers when considering the cost. These rows were decided to be removed from the data set. Usually, the cost is calculated for duration. Hence, any imputation techniques were not considered due to both variable values were acting as outliers in the above filtered data rows.

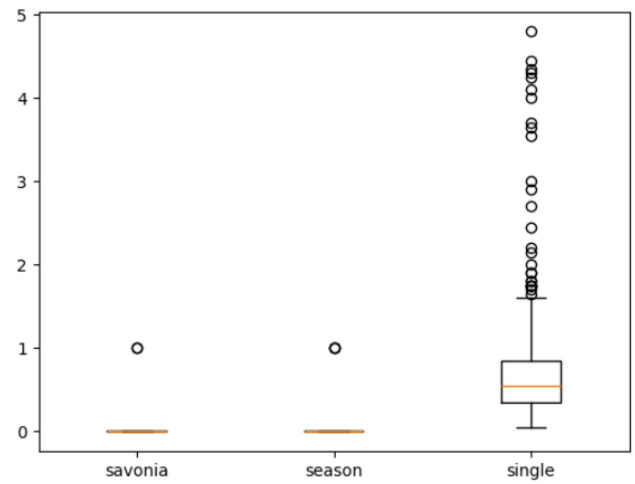
After removing the extreme outliers in the duration variable, the scatter plot looks as follows.



Cost

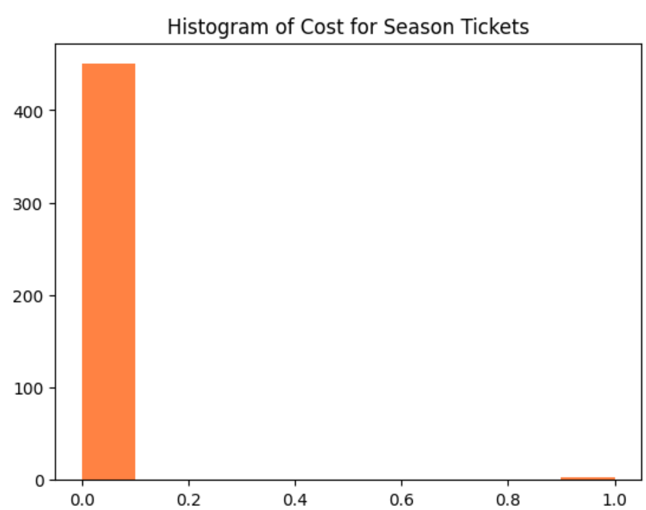
Created box plots for cost variable grouped by the ticket type.

```
#creating a box plot to explore 'cost' variable values grouped by ticket type

grouped_by_ticket = filtered_df.groupby('ticket')
plt.boxplot([group['cost'] for _, group in grouped_by_ticket], labels=grouped_by_ticket.groups.keys())
plt.show()
```



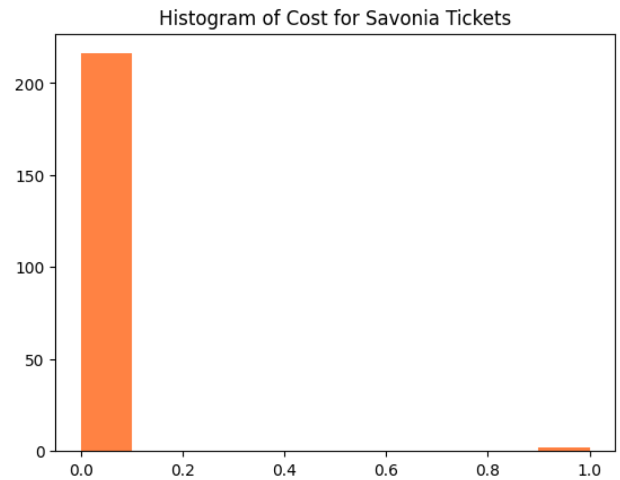
Creating a histogram of cost for filtered for season tickets as the box plots of cost for savonia and season tickets looks abnormal.

```
#Created a histogram of cost values for season ticket

plt.hist(filtered_df[filtered_df['ticket']=='season']['cost'],color='#ff8243')
plt.title('Histogram of Cost for Season Tickets')
plt.show()
```



Created a histogram of cost for filtered for savonia tickets.

```python
#Created a histogram of cost values for savonia ticket

plt.hist(filtered_df[filtered_df['ticket']=='savonia']['cost'],color='#ff8243')
plt.title('Histogram of Cost for Savonia Tickets')
plt.show()
```



Filtering out the data where the ticket type is not 'single' and the cost is greater than 0.

```python
#checking the outliers in savonia and season ticket types

filtered_df[(filtered_df['ticket'] != 'single') & (filtered_df['cost']>0)]
```

	ticket	cost	month	location_from	location_to	duration	distance	assistance	energy_used	energy_collected
251	season	1.0	6	KEILANKANTA	SATAMA	2021.0	7040	1	37.1	13.8
792	savonia	1.0	10	TORI	KAUPPAKATU	2011.0	4730	1	41.7	16.5
884	season	1.0	10	TASAVALLANKATU	TASAVALLANKATU	1918.0	7940	1	68.7	20.1
975	season	1.0	6	TORI	TORI	2660.0	4100	1	25.5	13.5
1569	savonia	1.0	10	TORI	TORI	1826.0	1590	0	0.0	2.7

As the cost >0 was only 5 out of all, it seems like an error. Replacing the cost value of the above data rows with 0.

Note: here in this specific case, 0 was assigned directly as the cost of the above because there's no variance in the data and the only possible value is 0. But if there are other values, median imputation can be used to replace these values.

```python
transformed_df.loc[(transformed_df['cost'] >0)&(transformed_df['ticket']!='single'), 'cost'] = 0
```

The following image shows the specific rows after applying the median imputation. The cost value is now replaced by 0.

	ticket	cost	month	location_from	location_to	duration	distance	assistance	energy_used	energy_collected
251	season	0.0	6	KEILANKANTA	SATAMA	2021.0	7040	1	37.1	13.8
792	savonia	0.0	10	TORI	KAUPPAKATU	2011.0	4730	1	41.7	16.5
884	season	0.0	10	TASAVALLANKATU	TASAVALLANKATU	1918.0	7940	1	68.7	20.1
975	season	0.0	6	TORI	TORI	2660.0	4100	1	25.5	13.5
1569	savonia	0.0	10	TORI	TORI	1826.0	1590	0	0.0	2.7

Energy_Used

In a trip, if the energy was used (energy_used>0) the electrical assistance should be on (assistance =1). It was assumed that there was no technical issue regarding recording the energy that was used. Filtering out the rows where assistance =0 and energy_used >0.

```
asstance_check_false = (transformed_df['assistance']==0) & (transformed_df['energy_used']>0)
transformed_df[asstance_check_false]
```

	ticket	cost	month	location_from	location_to	duration	distance	assistance	energy_used	energy_collected
231	single	0.10	7	TASAVALLANKATU	TASAVALLANKATU	61.0	1300	0	9.0	11.1
611	season	0.00	10	PIRTTI	PIRTTI	1795.0	1880	0	12.1	2.7
1372	savonia	0.00	10	TORI	TORI	8.0	190	0	1.5	0.6
1453	single	0.55	7	TORI	PUIJONLAAKSO	611.0	2590	0	25.4	3.3

Replacing assistance value to 1 in the above filtered rows.

```
transformed_df.loc[(transformed_df['assistance']==0) & (transformed_df['energy_used']>0), 'assistance']=1
```

If the energy_used is 0 during a trip, it means that the electrical assistance was off. Filtering out the rows where the energy_used was 0 but the assistance is showing as 1. In the results there were 15.

```
asstance_check_true = (transformed_df['assistance']==1) & (transformed_df['energy_used']==0)
transformed_df[asstance_check_true]
```

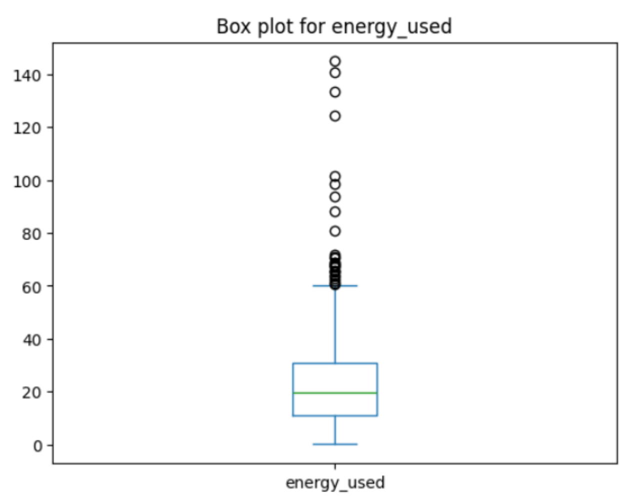
	ticket	cost	month	location_from	location_to	duration	distance	assistance	energy_used	energy_collected
193	single	0.10	7	NEULAMÄKI	NEULAMÄKI	101.0	10	1	0.0	0.0
254	single	0.10	7	PUIJONLAAKSO	PUIJONLAAKSO	66.0	30	1	0.0	0.0
321	savonia	0.00	9	TORI	TORI	112.0	130	1	0.0	0.0
375	single	0.05	7	TORI	TORI	57.0	10	1	0.0	0.0
507	single	0.05	6	SATAMA	SATAMA	59.0	80	1	0.0	0.0
940	season	0.00	6	TASAVALLANKATU	TASAVALLANKATU	67.0	10	1	0.0	0.0
956	single	1.05	6	TORI	PUIJONLAAKSO	1214.0	10	1	0.0	0.0
1027	season	0.00	9	KEILANKANTA	KEILANKANTA	74.0	10	1	0.0	0.0
1049	season	0.00	6	TORI	TORI	197.0	60	1	0.0	0.0
1102	single	1.20	9	KEILANKANTA	TASAVALLANKATU	1408.0	4400	1	0.0	0.0
1524	single	0.10	8	KAUPPAKATU	KAUPPAKATU	91.0	20	1	0.0	0.0
1709	single	0.55	4	TASAVALLANKATU	KAUPPAKATU	633.0	1610	1	0.0	3.3
1740	single	0.10	6	TASAVALLANKATU	TASAVALLANKATU	84.0	50	1	0.0	0.0
1747	season	0.00	6	NEULAMÄKI	NEULAMÄKI	111.0	10	1	0.0	0.0
1753	season	0.00	8	TORI	TORI	113.0	210	1	0.0	0.3

Replacing the assistance value of the above data rows with 0.

```
transformed_df.loc[(transformed_df['assistance']==1) & (transformed_df['energy_used']==0), 'assistance']=0
```

A box plot was created to check for any outliers in the energy_used data and it was decided to keep them in the analysis as they seem to be valid and important for the analysis.

```
#creating a box plot for energy_used

transformed_df[transformed_df['assistance']==1]['energy_used'].plot.box()
plt.title('Box plot for energy_used')
plt.show()
```



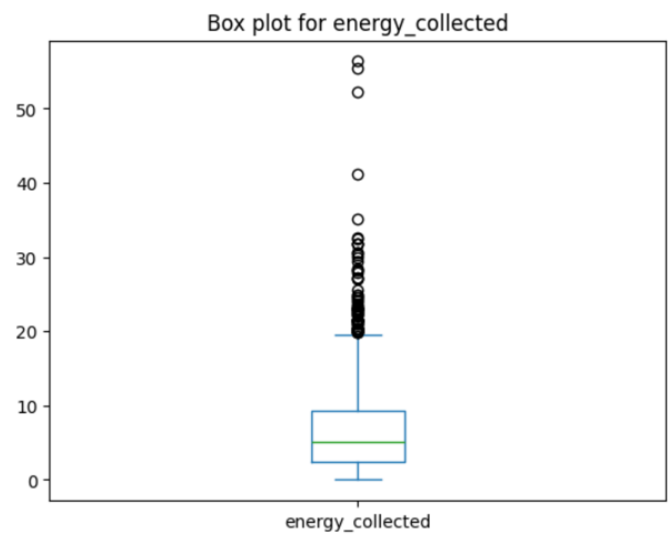
```
transformed_df[transformed_df['energy_used']>120]
```

	ticket	cost	month	location_from	location_to	duration	distance	assistance	energy_used	energy_collected
555	single	3.65	8	KEILANKANTA	KAUPPAKATU	4345.0	14240	1	140.6	31.8
705	single	3.55	6	PUIJONLAAKSO	PUIJONLAAKSO	4207.0	17410	1	144.9	56.4
1202	single	4.10	8	NEULAMÄKI	NEULAMÄKI	4911.0	20770	1	124.2	52.2
1759	single	4.30	5	KEILANKANTA	KEILANKANTA	5136.0	18720	1	133.4	22.2

Energy_Collected

A box plot was created to check for any outliers in the energy_collected data. After examining the rows, it was decided to keep them in the analysis as they seem to be valid and important.

```
#creating a box plot for energy_collected

transformed_df['energy_collected'].plot.box()
plt.title('Box plot for energy_collected')
plt.show()
```




```
transformed_df[transformed_df['energy_collected']>40]
```

	ticket	cost	month	location_from	location_to	duration	distance	assistance	energy_used	energy_collected
705	single	3.55	6	PUIJONLAAKSO	PUIJONLAAKSO	4207.0	17410	1	144.9	56.4
1202	single	4.10	8	NEULAMÄKI	NEULAMÄKI	4911.0	20770	1	124.2	52.2
1571	single	1.50	7	NEULAMÄKI	SATAMA	1782.0	6510	1	38.1	41.1
1649	single	4.00	5	TORI	TORI	4793.0	11200	1	98.3	55.5

Duplicates

I have checked the data set for possible duplicates. And I have found 5 pairs of data rows. since there are no user data and timestamp user data cannot be identified as duplicates. They can be actual observations.

```
duplicates = transformed_df[transformed_df.duplicated(keep=False)]
duplicates.sort_values(by='duration')
```

	ticket	cost	month	location_from	location_to	duration	distance	assistance	energy_used	energy_collected
228	savonia	0.00	10	TORI	KAUPPAKATU	111.0	390	0	0.0	0.6
1392	savonia	0.00	10	TORI	KAUPPAKATU	111.0	390	0	0.0	0.6
460	savonia	0.00	9	NEULAMÄKI	MICROTEKNIA	250.0	1400	0	0.0	6.9
887	savonia	0.00	9	NEULAMÄKI	MICROTEKNIA	250.0	1400	0	0.0	6.9
445	season	0.00	5	KAUPPAKATU	TASAVALLANKATU	351.0	1610	1	11.9	2.7
626	season	0.00	5	KAUPPAKATU	TASAVALLANKATU	351.0	1610	1	11.9	2.7
1056	single	0.35	7	PUIJONLAAKSO	MICROTEKNIA	410.0	2140	1	12.5	20.4
1623	single	0.35	7	PUIJONLAAKSO	MICROTEKNIA	410.0	2140	1	12.5	20.4
737	single	0.75	7	SATAMA	PUIJONLAAKSO	887.0	3360	1	45.0	5.7
823	single	0.75	7	SATAMA	PUIJONLAAKSO	887.0	3360	1	45.0	5.7

After exploring the variables, filtering out irrelevant records and processing invalid values, the observation count in the dataset is 1475.

Data Exploration

Number of Trips Made by Each Ticket Type

First, we will find out the number of trips made by each type of ticket.

```
# Number of trips made

transformed_df.groupby('ticket')['ticket'].count().to_frame().rename(columns={'ticket': 'Number of Trips'})
```

	Number of Trips
ticket	
savonia	218
season	453
single	804

Looking at the results, it can be concluded that most of the trips were made by the users who have purchased a single ticket type. About 54% of the trips were made by single ticket users while about 31% were made by users who have season tickets and about 15% by the savonia ticket holders.

Therefore we can assume that most users prefer single tickets over the season tickets or savonia tickets.

Total Distance and Time Travelled by Each Ticket Type

In this section we will calculate the summation of distance and duration grouped by each ticket type.

```
# Total distance traveled, time and total fees paid

transformed_df.groupby('ticket')[['distance','duration']].sum()
```

	distance	duration
ticket		
savonia	494630	137753.0
season	1311590	314730.0
single	2551610	633204.0

When it comes to the highest total travel distance and highest total trip duration made by the users, we can see that the single ticket users, season ticket users and savonia ticket users take the 1st, 2nd and 3rd places respectively.

If we look at the mean values of distance and durations,

```
# Mean distance traveled, time and total fees paid

transformed_df.groupby('ticket')[['distance','duration']].mean()
```

	distance	duration
ticket		
savonia	2268.944954	631.894495
season	2895.342163	694.768212
single	3173.644279	787.567164

It seems that the savonia students usually rent bikes to travel shorter distances and durations.

Total Amount of Fees Paid by Each Ticket Type

```
# Total distance traveled, time and total fees paid

transformed_df.groupby('ticket')[['cost']].sum()
```

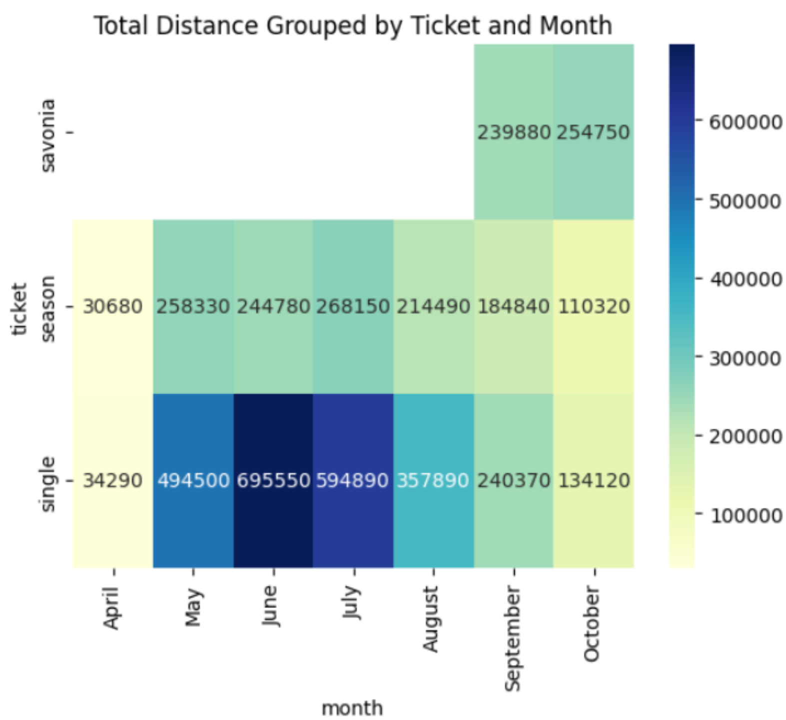
	cost
ticket	
savonia	0.00
season	0.00
single	548.25

When it comes to ‘cost’, 548.25 euros were paid as the total fee by the single ticket users. It was assumed that the season and savonia ticket holders fee are not included in the data set as it must be a one-time payment for a specific period because the majority of the data contained 0 as the cost.

Monthly Rental Activity in Terms of The Total Distance Travelled

To explore the monthly rental activities considering the total travelled distance, the total distance was calculated grouping by the ticket type and month. Created a heatmap to visualize the data.

		distance
ticket	**month**	
savonia	**September**	239880
	October	254750
season	**April**	30680
	May	258330
	June	244780
	July	268150
	August	214490
	September	184840
	October	110320
single	**April**	34290
	May	494500
	June	695550
	July	594890
	August	357890
	September	240370
	October	134120



The code use to create the heatmap:

```
#create pivot table

pivot_table = result.pivot_table(values='distance', index='ticket', columns='month')
```

```
#pivot_table
```

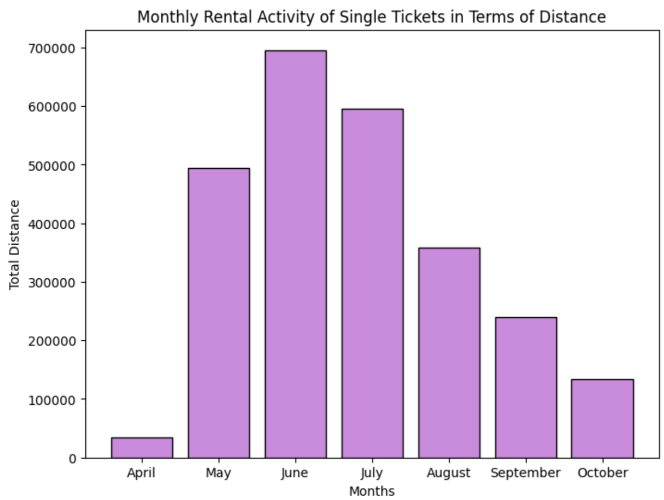
```
#re-order months

month_order = ['April','May','June','July','August','September','October']
pivot_table = pivot_table.reindex(columns=month_order)
```

```
pivot_table
```

month	April	May	June	July	August	September	October
ticket							
savonia	NaN	NaN	NaN	NaN	NaN	239880.0	254750.0
season	30680.0	258330.0	244780.0	268150.0	214490.0	184840.0	110320.0
single	34290.0	494500.0	695550.0	594890.0	357890.0	240370.0	134120.0

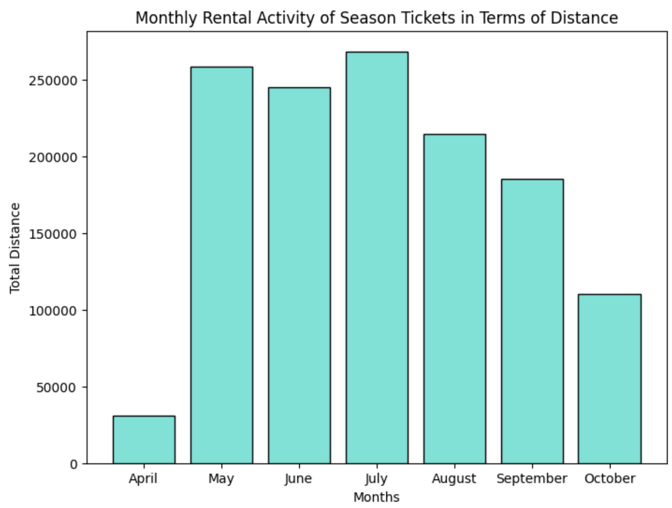
Looking at the heatmap, it can be clearly seen out of all, the highest monthly rental in terms of total travel distance was made in June by the single ticket type users. The monthly rental activity varies significantly in single ticket users.

```
# Create a bar plot for the Monthly Rental Activity of Single Ticket in Terms of Distance
months=['April','May','June','July','August','September','October']
plt.figure(figsize=(8, 6))

plt.bar(months, result.loc['single','distance'],edgecolor='black', color='#c98bdb')
plt.xlabel('Months')
plt.ylabel('Total Distance')
plt.title('Monthly Rental Activity of Single Tickets in Terms of Distance')

plt.show()
```



In the Season ticket users,

```
# Create a bar plot for the Monthly Rental Activity of Single Ticket in Terms of Distance
months=['April','May','June','July','August','September','October']
plt.figure(figsize=(8, 6))

plt.bar(months, result.loc['season','distance'],edgecolor='black', color='#82e1d7')
plt.xlabel('Months')
plt.ylabel('Total Distance')
plt.title('Monthly Rental Activity of Season Tickets in Terms of Distance')

plt.show()
```



It can be observed that most bikes were used during the summer period. So, it can be suggested that more bikes are needed in this duration.

On the other hand, the savonia ticket holders haven't made any rentals during the April-August period. But the total trip distances have risen to a moderate amount in September and have increased more in October months. It can be assumed that this is due to the commencement of the Autumn intake of the universities.

It can be suggested that starting from September the availability of the bikes should be increased near universities and student housing areas.

Stations That Have the Highest Total Deficit and Highest Total Surplus of Bikes

To find out the stations that have the highest deficit and surplus, The number of bikes leaving the stations and the number of bikes arriving at the stations were calculated as follows.

```
trips_started=transformed_df.groupby(['location_from'])['ticket'].count().rename('Number of Trips Started')
trips_ended=transformed_df.groupby(['location_to'])['ticket'].count().rename('Number of Trips Ended')
```

```
trips_started
```

```
location_from
KAUPPAKATU       194
KEILANKANTA      130
KYS               80
MICROTEKNIA      157
NEULAMÄKI        144
PIRTTI            25
PUIJONLAAKSO     134
SATAMA           167
TASAVALLANKATU   111
TORI             333
Name: Number of Trips Started, dtype: int64
```

```
trips_ended
```

```
location_to
KAUPPAKATU       236
KEILANKANTA      120
KYS               63
MICROTEKNIA      149
NEULAMÄKI        144
PIRTTI            26
PUIJONLAAKSO     164
SATAMA           128
TASAVALLANKATU   125
TORI             320
Name: Number of Trips Ended, dtype: int64
```

The three stations that have the highest total deficit of bikes are,

location_to	Difference
KAUPPAKATU	42
KEILANKANTA	-10
KYS	-17
MICROTEKNIA	-8
NEULAMÄKI	0
PIRTTI	1
PUIJONLAAKSO	30
SATAMA	-39
TASAVALLANKATU	14
TORI	-13

- Satama
- KYS
- Tori

The three stations that have the highest total surplus of bikes are,

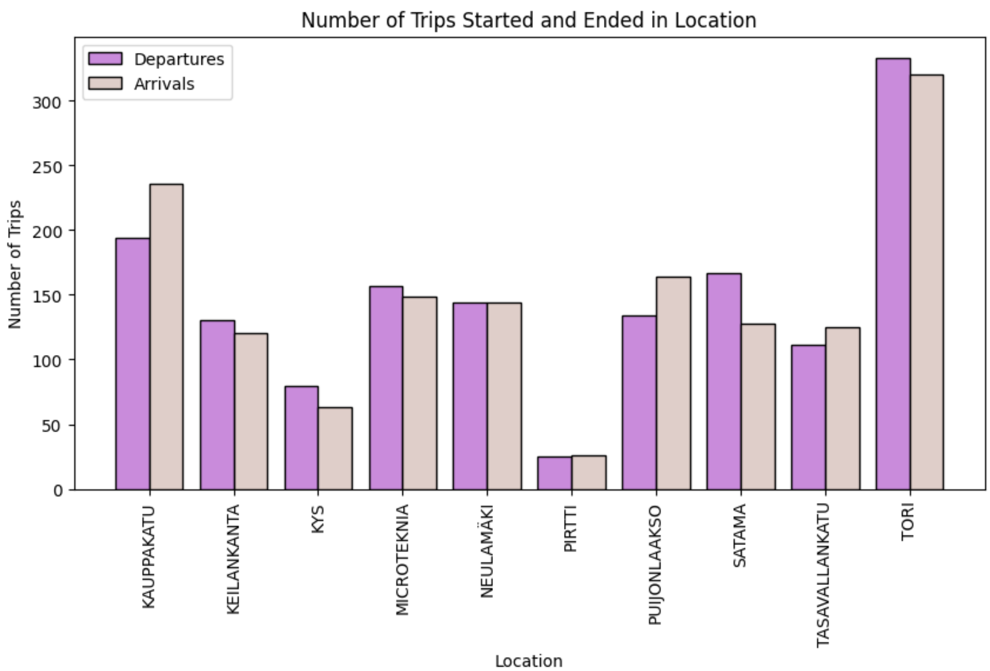
- Kauppakatu
- Puijonlaakso
- Tasavallankatu

The highest congregation of bikes can be seen in Tori while the lowest is in Pirtti. In Tori, being the most popular station, the amount of bikes can be increased because there's a considerable deficit of bikes.

It can be suggested that some amount of bikes can be relocated from Kauppakatu to Satama.

```
plt.figure(figsize=(10, 5))

r= np.arange(10)

plt.bar(r, trips_started,0.4,color='#c98bdb',edgecolor='black')
plt.bar(r+0.4, trips_ended,0.4,color='#dfcec9',edgecolor='black')

plt.xlabel('Location')
plt.ylabel('Number of Trips')
plt.title('Number of Trips Started and Ended in Location')
plt.legend(['Departures', 'Arrivals'])

plt.xticks(r+0.2, trips_started.index,rotation=90)

plt.show()
```

The total deficit and surplus can be seen clearly in the following graph.



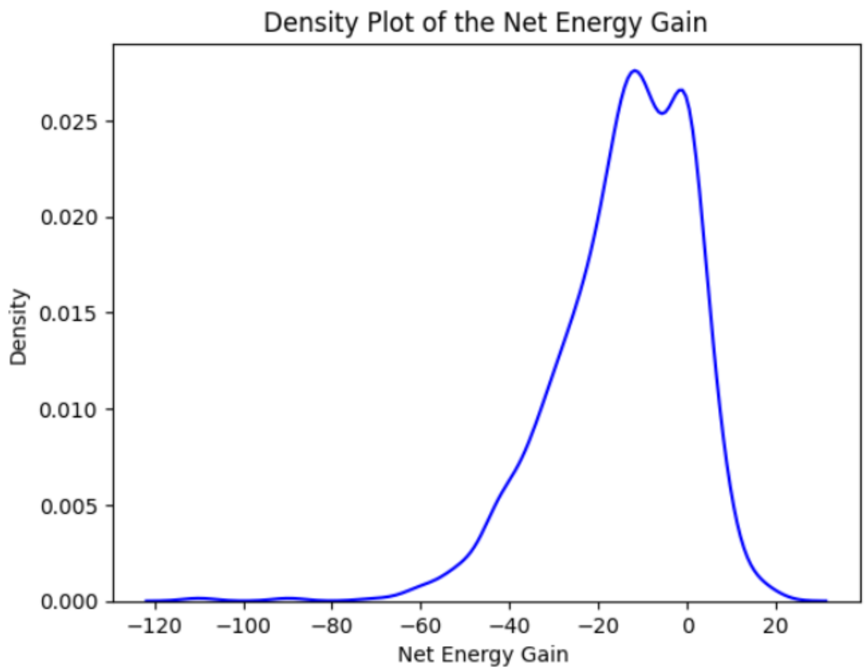
Net Energy Gain During Trips

```
#Create new column 'net_energy_gain'
transformed_df['net_energy_gain']=transformed_df['energy_collected']-transformed_df['energy_used']
```

```
transformed_df.head(5)
```

	ticket	cost	month	location_from	location_to	duration	distance	assistance	energy_used	energy_collected	net_energy_gain
0	single	0.35	9	MICROTEKNIA	PUIJONLAAKSO	411.0	2150	1	19.0	2.7	-16.3
1	single	1.20	5	SATAMA	KEILANKANTA	1411.0	7130	1	53.8	15.3	-38.5
2	savonia	0.00	9	TASAVALLANKATU	NEULAMÄKI	1308.0	5420	1	43.0	9.9	-33.1
3	savonia	0.00	10	TORI	KAUPPAKATU	1036.0	1180	1	6.5	2.1	-4.4
4	single	0.30	9	TORI	TORI	319.0	1120	1	13.7	1.2	-12.5

```
sns.kdeplot(transformed_df['net_energy_gain'], color='blue')
plt.title('Density Plot of the Net Energy Gain')
plt.xlabel('Net Energy Gain ')
plt.ylabel('Density')

plt.show()
```



By looking at the distribution of data, it is apparent that the net energy gain tends to be low during most of the trips. Net energy loss is greater.

On average, bikes tend to lose 14 watt-hours during a trip.

```
transformed_df['net_energy_gain'].mean()
```

```
-14.08291525423729
```

Hypothesis Testing

The confidence level of the below statistical tests is considered as 95%. (Alpha = 0.05)

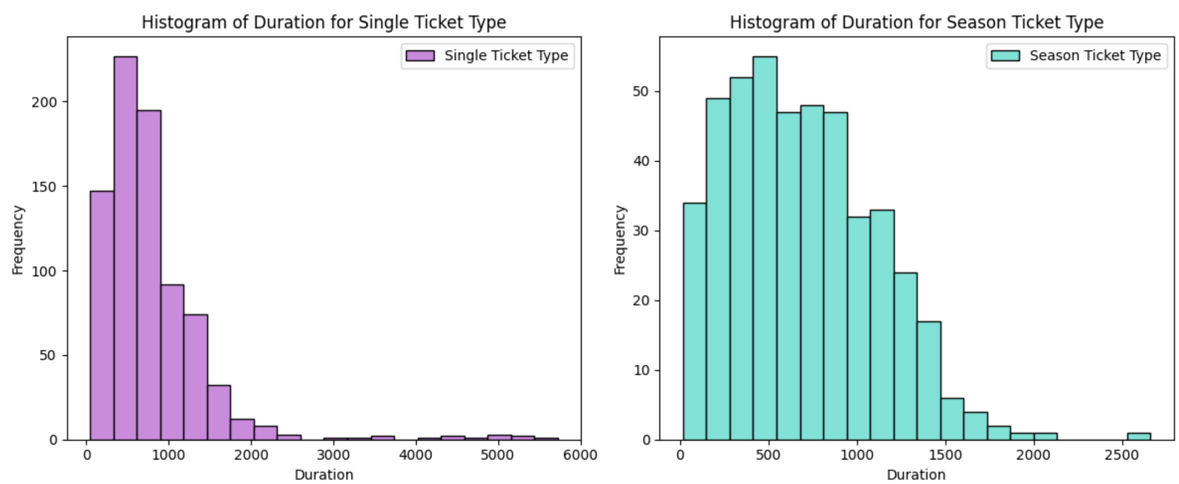
Hypothesis- Trip durations tend to be different between the customers who have a season ticket and the customers who buy single tickets.

Creating two samples of duration data by season and single ticket types.

```
group_single_v1 = df_version1[df_version1['ticket'] == 'single']['duration']
group_season_v1 = df_version1[df_version1['ticket'] == 'season']['duration']
```

Visualizing the two data samples using histograms.

```
plt.figure(figsize=(12, 5))
plt.subplot(1, 2, 1)

#Create Histogram for duration for single ticket type

plt.hist(group_single, bins=20, label='Single Ticket Type',edgecolor='black', color='#c98bdb')

plt.legend()
plt.xlabel('Duration')
plt.ylabel('Frequency')
plt.title('Histogram of Duration for Single Ticket Type')

plt.subplot(1, 2, 2)

#Create Histogram for duration for season ticket type

plt.hist(group_season, bins=20, label='Season Ticket Type',edgecolor='black', color='#82e1d7')

plt.legend()
plt.xlabel('Duration')
plt.ylabel('Frequency')
plt.title('Histogram of Duration for Season Ticket Type')

plt.tight_layout()
plt.show()
```



Checking for normality of the two data sets using Shapiro-Wilk test.

```
print(stat.shapiro(group_season_v1).pvalue, end="\n\n")
```

```
2.4213724358901345e-09
```

```
print(stat.shapiro(group_single_v1).pvalue, end="\n\n")
```

```
3.4303258352568613e-35
```

The p-values returned from the tests for both samples are smaller than 0.05. There's no evidence to prove that the data is normal.

In this scenario, it is asked to compare two groups within a categorical variable in terms of a continues variable. As the data is not normal and unpaired, applying the Mann–Whitney U test to find out whether there's statistical evidence to prove that there's a difference between the two data sets.

The null hypothesis (H_0): There is no difference between the trip durations of the two populations.

The alternative hypothesis (H_A): The trip durations are different between the two populations.

Performing test on the data,

```
print(stat.mannwhitneyu(group_season, group_single, use_continuity=True).pvalue, end="\n\n")
```

```
0.2885827017890521
```

P- value is approximately 0.289 which is greater than 0.05. This suggests that there is not enough evidence to reject the null hypothesis. In other words, the test did not find a statistically significant difference between the two groups.

This test doesn't provide enough evidence to prove that the trip durations differ between the customers who have a season ticket and the customers who travel on single tickets.

It seems that customers are utilizing both single and seasonal tickets equally according to their needs.

Hypothesis- Travel distance positively correlates with the average rate at which electricity is consumed during the trip.

Energy consumption rate = Energy used in each trip / duration

Added new column energy_consumption_rate in the data frame.

```
transformed_df['energy_consumption_rate']=transformed_df['energy_used']/transformed_df['duration']
```

```
transformed_df.head(5)
```

	ticket	cost	month	location_from	location_to	duration	distance	assistance	energy_used	energy_collected	net_energy_gain	energy_consumption_rate
0	single	0.35	9	MICROTEKNIA	PUIJONLAAKSO	411.0	2150	1	19.0	2.7	-16.3	0.046229
1	single	1.20	5	SATAMA	KEILANKANTA	1411.0	7130	1	53.8	15.3	-38.5	0.038129
2	savonia	0.00	9	TASAVALLANKATU	NEULAMÄKI	1308.0	5420	1	43.0	9.9	-33.1	0.032875
3	savonia	0.00	10	TORI	KAUPPAKATU	1036.0	1180	1	6.5	2.1	-4.4	0.006274
4	single	0.30	9	TORI	TORI	319.0	1120	1	13.7	1.2	-12.5	0.042947

Check for normality of data using the Shapiro-Wilk test.

```
print(stat.shapiro(transformed_df['distance']).pvalue, end="\n\n")
```

```
1.254139330996114e-32
```

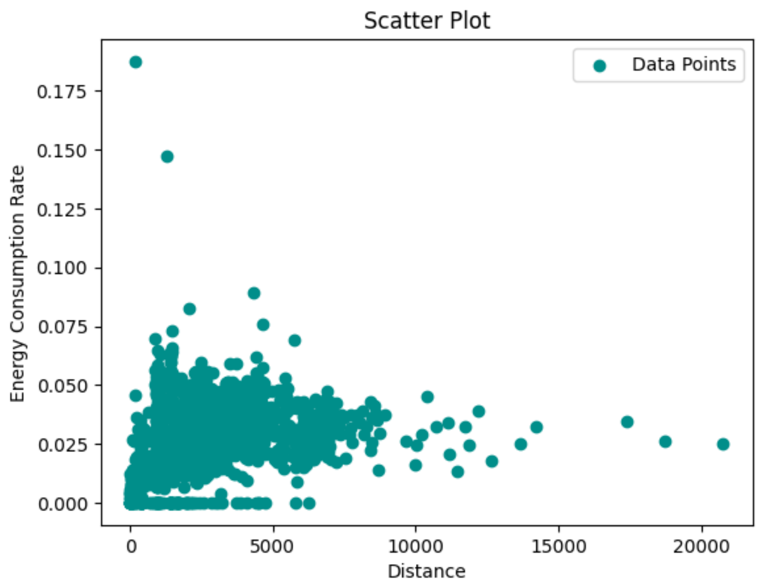
```
print(stat.shapiro(transformed_df['energy_consumption_rate']).pvalue, end="\n\n")
```

```
7.391108005875698e-25
```

The data seems not normal.

Creating a scatter plot to visualize the relationship between the energy consumption rate and the distance.

```
plt.scatter(transformed_df['distance'], transformed_df['energy_consumption_rate'], label='Data Points', c='#008e8a', marker='o')

plt.xlabel('Distance')
plt.ylabel('Energy Consumption Rate')
plt.title('Scatter Plot')
plt.legend()

plt.show()
```



The scatter plot doesn't show a strong positive linear relationship between the two variables. But the data also looks less scattered and highly congested. So it is difficult to conclude the relationship by looking at the plot.

The null hypothesis (H_0): There is no statistically significant correlation between duration and energy consumption rate (correlation coefficient ρ = 0)

The alternative hypothesis (H_A): There is a statistically significant correlation between duration and energy consumption rate (correlation coefficient ρ ≠ 0)

Applying Spearman correlation test to find out the correlation coefficient between the distance and energy consumption rate.

```
print(tuple(stat.spearmanr(transformed_df['distance'], transformed_df['energy_consumption_rate'])), end="\n\n")
```

```
(0.3052330582858624, 3.540820213825516e-33)
```

The correlation coefficient is approximately 0.305 which indicates a positive correlation between the two variables. But as it is closer to 0 than 1, it is not considered as a strong correlation.

The test has also returned a p-value which is smaller than 0.05. By which we can reject the null hypothesis and conclude that the above correlation is statically significant.

This test provides statistical evidence to claim that the travel distance maintains a slightly positive correlation with the electricity consumption rate during trips.

We can make a conclusion that the customers are likely to use electrical assistance during long distance trips.

Hypothesis- savonia ticket type differs from the others with respect to how often the electric assistance is used.

In this scenario we need to compare ticket type and electrical assistance, which are two categorical variables.

As the hypothesis is about savonia ticket types and other ticket types, the ticket type is now separated to two values which are ‘savonia’ and ‘other’. This new column ‘new_ticket_type’ was added to the data frame.

```
transformed_df['new_ticket_type'] = np.where(transformed_df['ticket'] == 'savonia', 'savonia', 'other')
transformed_df.head(5)
```

	ticket	cost	month	location_from	location_to	duration	distance	assistance	energy_used	energy_collected	net_energy_gain	energy_consumption_rate	new_ticket_type
0	single	0.35	9	MICROTEKNIA	PUIJONLAAKSO	411.0	2150	1	19.0	2.7	-16.3	0.046229	other
1	single	1.20	5	SATAMA	KEILANKANTA	1411.0	7130	1	53.8	15.3	-38.5	0.038129	other
2	savonia	0.00	9	TASAVALLANKATU	NEULAMÄKI	1308.0	5420	1	43.0	9.9	-33.1	0.032875	savonia
3	savonia	0.00	10	TORI	KAUPPAKATU	1036.0	1180	1	6.5	2.1	-4.4	0.006274	savonia
4	single	0.30	9	TORI	TORI	319.0	1120	1	13.7	1.2	-12.5	0.042947	other

First, we will check whether there’s a relationship between the two variables which are the new_ticket_type and assistance.

The null hypothesis (H_0): There’s no association between ticket type and the use of electrical assistance.

The alternative hypothesis (H_A): There’s an association between ticket type and the use of electrical assistance.

```
contingency_table = pd.crosstab(transformed_df['new_ticket_type'], transformed_df['assistance'])
```

```
print(contingency_table)
```

assistance	0	1
new_ticket_type		
other	76	1181
savonia	30	188

```
p_value= stat.chi2_contingency(contingency_table)[1]
chi2, p, dof, expected =  stat.chi2_contingency(contingency_table)

print("P-value:", p)
```

```
P-value: 8.501342153935647e-05
```

There’s statistical evidence to reject the null hypothesis. Which clearly shows that there’s an association between the use of electrical assistance and the ticket type.

Now, let’s look at the expected frequencies returned from the chi squared test.

```python
print("Expected Frequencies Table:")

pd.DataFrame(expected, index=['other','savonia'])
```

	0	1
other	90.333559	1166.666441
savonia	15.666441	202.333559

By comparing the above table with the contingency table, we can see that the observed frequency for the savonia students using the electrical assistance is smaller than expected or savonia students not using the electrical assistance is higher than expected. The observed frequency for the other ticket types is for using the electrical assistance is a bit higher than expected.

In other words, there is statistical evidence to claim that the savonia ticket type differs from the others with respect to how often the electric assistance is used.

As the savonia students are less likely use the electrical assistance, it would be a positive step to make these bikes available to them in stations near to universities and accommodation villages.

Future directions

Choosing (A).

Hypothesis: Customers departing from Tori are likely to use electrical assistance than the customers who are departing from other stations.

Assuming that Tori is a city square with the likelihood of a lot of workplaces, as the highest congregation of bikes are centered around it, we can assume that people tend to use electrical assistance for efficiency and conserve their energy at the end of a long day.

To find evidence to claim this, we can use chi-squared test to find out whether the use of electrical assistance differs from the departing stations (location_from). After creating a contingency table with the frequencies of assistance on and off for each station, we can perform chi-squared on the data and obtain the p-value to find out whether there's a significant difference among the locations. By exploring the expected frequencies and current observations, we can find out whether the hypothesis is valid.

If we can find statistical evidence to prove this hypothesis, more electrically charged bikes can be made available in Tori. This also can be used for marketing purposes on electric bikes as the cities are equipped with various platforms for marketing than other places.